

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

April 2018

EE405 Power System Protection

Date: 04/04/2018, Wednesday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

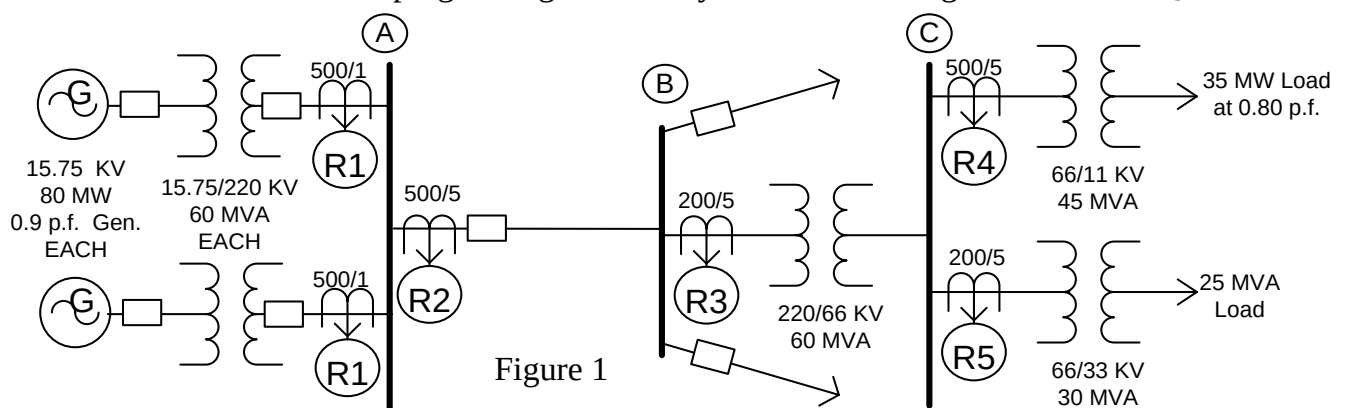
SECTION – I

Q – 1 Fill in the blanks.

[07]

- i. Time setting range of ground relay is _____ that of phase relay.
- ii. The ideal value of drop-off to pick-up ratio is _____.
- iii. The phase relay must be reach up to the end of next protection zone for the _____ fault.
- iv. Buchholz relay is provided for protection of _____.
- v. For parallel feeder protection, the _____ relay is used at load end.
- vi. Disc of IDMT relay is generally made up of _____.
- vii. The operating time of an IDMT relay corresponding to PSM=1 and TMS=1 will be _____.

Q – 2 A Figure 1 shows a single-line diagram of a portion of a power-system network. The relays shown are standard IDMT relays (setting range= 50 to 200% of CT secondary rating). The three phase short circuit current of bus A, B and C are 3000 Amp, 2500 Amp and 6000 Amp respectively. TMS of R_5 and R_4 are set as 0.1 each. Determine the plug setting of all relays and time setting of R_1, R_2 and R_3 . [08]



B Give Reason: Inverse definite minimum time over-current relays give better discrimination than inverse-time over-current relays. [03]

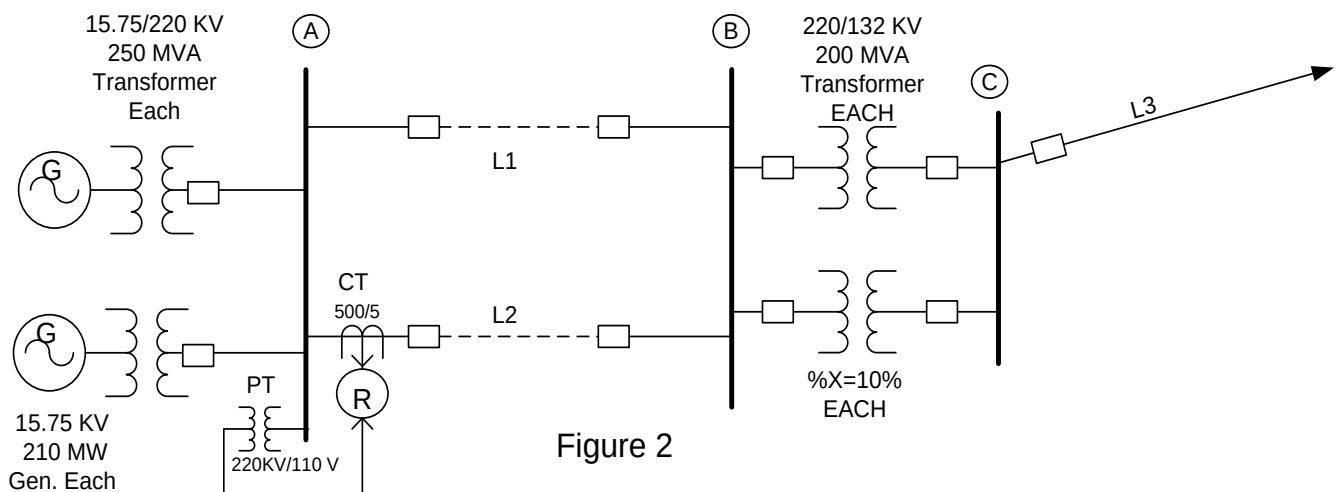
C Give Reason: Dead zone is present in directional relay. [03]

Q – 3 Attempt any TWO

- A** The part of the power system is shown in figure 2. Decide the three zone settings of a mho distance relay R having characteristic angle of 65° . First zone covers 80% of transmission line L_2 , second zone reach is up to bus C and third zone covers 50% of transmission line after bus C. [07]

Required data is as follows:

- (i) 220 KV Lines L_1 and L_2
Impedance: $0.04 + j 0.2$ ohm / km / ph. Line length: 200 Km.
(ii) 132 KV line L_3 :
 L_3 : Impedance: $0.05 + j 0.15$ ohm / Km / ph. Line length: 100 Km



- B** Draw the block diagram of carrier blocking scheme used with distance relay for transmission line protection and explain it. [07]
- C** Explain with mathematical proof: How the phasors are estimated for relaying quantity. [07]

SECTION – II

- Q – 4.** I. Give reason: Though the stator Earth fault is an internal fault, it is not detected by generator over-all differential protection. [02]
- II. With proper vector diagram explain, how the NPS relay responds only to the negative phase sequence components only. [05]

Q – 5. Attempt any TWO

- A** With neat and clean diagram, show that how distance relay can be saved from mal-operation during blowing of fuse in the PT circuit. [07]
- B** With neat and clean diagram, explain the generator stator inter-turn fault protection. [07]

- C** Following are the details of 3 phase, 750 HP, 6 KV Induction motor. **[07]**
- i. Rated p.f.:0.85
 - ii. Efficiency: 80%
 - iii. Starting time: 6 seconds
 - iv. Starting Current: 5.5 times the rated current
 - v. CT ratio: 100/1 Amp.
 - vi. Motor can able to take 110% of rated current continuously.

Relay: Rating =1 amp.

- a) Thermal O/L Unit: 70%-130% of 1 Amp in step of 10%
- b) Instantaneous O/C Unit: 400%-1500% of 1 Amp in step of 50%
- c) NPS Unit: 100% -600% of 1 Amp in steps of 50%
- d) Stalling Unit:
 - Current Setting Range: 90-600% of 1 Amp in step of 30%
 - Time Setting Range: 4-30 seconds.
 - (Safe stalling time of I.M. Hot=13 sec; Cold=21 Sec.)
- e) Under Voltage Relay: Range 70-100% of 110 V. (P.T. 6 KV/110V)

Suggest the setting of above mentioned relay for proper protection of I.M.

Q – 6. Attempt any TWO

- A** Enlist and explain the faults in auxiliary equipment of power transformer **[07]**
- B** List out and explain the problems encountered in Protective CT. **[07]**
- C** Enlist the type tests that are performed on the protective relays. Explain each of them. **[07]**
